

Unix/Linux Operating Systems – Make-up Exam

Duration: 2 hours

Total Points: 20

Use of phones, documents, or any form of cheating is strictly prohibited. Any student caught cheating will receive a grade of 0.

Name:

Group:

Section A – Theory & Concepts (8 pts)

Q1. (2 pts)

For each of the following commands or keywords, briefly explain what it does and give an example:

a) alias

b) export

c) ssh

d) scp

Q2. (1.5pt)

What is rsync. Write the command to use it and explain the main difference with scp

Q3. (1.5 pt)

Explain how to:

a) Run a command in the background

b) List background jobs

c) Resume a background job in the foreground

Q4. (2 pts)

You have a shell script in your home directory and you want to run it from anywhere like a regular command. What should you do? The script is called `myscript.sh`

Q5. (1 pt)

List four different modes in Vim and give the command or key you use to enter each mode from Normal mode.

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Section B – Practical & Scripting (12 pts)

Q6. (1 pt)

Write a command that creates a folder named `archives`, compresses all `.txt` files in the current directory into a “data.tar.gz” archive, and moves the archive into “archives/”.

Q7. (2 pt)

You want a message 'Hello \$USER' to appear at each session start using systemd:

a) Write script ~/hello.sh

b) Create service file hello.service

c) Enable the service

Q8. (2 pts)

Write shell commands to:

a) Create a user named “user”

b) Create a group named “unix”

c) Add `user` to the “unix” group

d) Set folder “class” so only “trainee” can read and write to it

Q9. (2 pts) Write a script “bmi_calculator.sh” that :

- Takes weight (kg) and height (m) as command-line arguments.
Example: `./bmi_calculator.sh 70 1.75`
- Computes the BMI using the formula:

$$\text{BMI} = \frac{\text{weight}}{\text{height}^2}$$

- Displays:
 - “Underweight” if $\text{BMI} < 18.5$
 - “Normal weight” if $\text{BMI} \geq 18.5$ and < 25
 - “Overweight” if $\text{BMI} \geq 25$ and < 30
 - “Obese” if $\text{BMI} \geq 30$

Prints both the BMI (rounded to 2 decimals) and the category.

Q10. (2 pts)

a) What is the default mode permission for a file? Explain the different numbers.

b) What will be the permission a file if you run umask 077 before?

Q11 Linux Networking (3pts, 6*0.5pts)

- Name the command you can use to list all network interfaces on a Linux system.
- How do you check the IPv4 address of your primary interface?
- Which command would you use to assign a static IP address to a network interface?
- What is the difference between a static IP and a dynamic IP?
- Which command can you use to check if you can reach google.com from your computer?
- How can you check which ports are currently listening on your system?